ARRIVAL-MNEMO

Persistent Memory Architecture for Cross-Architecture AI Coordination

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Repository: `https://github.com/mifkilla/Arrival-Protocol`

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1. Abstract

The Arrival Protocol enables cross-architecture AI-to-AI coordination through structured semantic @-atoms (DEUS.PROTOCOL v0.5), validated across 2,200+ experiments. However, a critical limitation persists: all coordination knowledge is lost between sessions. We present Arrival-Mnemo, a persistent memory architecture extending the protocol with four memory layers (Episodic, Procedural, Semantic, Meta), a `@MEMORY.*` atom family, and MEMORY-CRDT v1.1 — a formal specification for cross-temporal memory synchronization using the same Care Resolve mathematics that powers real-time dialogue convergence.

We conducted three experimental phases. Phase 14 reveals a **negative result**: naive keyword-based global memory injection DEGRADES accuracy by -5.7 pp (McNemar $p = 0.724$) on GPQA Diamond, discovering the *hypercorrection effect* in LLM ensembles — a cognitive bias previously documented only in humans. Phase 15 introduces Care-ALERT, a gated real-time intervention system that restores baseline accuracy while achieving the first statistically significant improvement in all Arrival experiments: Care Resolve is significantly higher than under global injection (Mann–Whitney $p = 0.042$).

Key findings: (1) memory creation via auto-logging is solved; (2) global pre-session injection triggers hypercorrection; (3) gated real-time intervention avoids hypercorrection while improving consensus quality; (4) specific memory helps but generic memory hurts AI coordination.

Keywords: AI coordination, persistent memory, CRDT, semantic atoms, multi-agent systems, hypercorrection, CARE-ALERT

2. Introduction

2.1 The Amnesia Problem

Modern AI coordination protocols operate in a fundamentally amnesiac mode. When a multi-agent session ends, all accumulated knowledge vanishes. The agents retain no recollection of which strategies worked, which information sources proved reliable, or how their peers typically behave. Every new session starts from absolute zero.

The Arrival Protocol [1] demonstrates that cross-architecture AI agents can achieve structured coordination through semantic @-atoms, with results validated across 2,200+ experiments, 23 phases, and 17 LLM architectures. But this coordination knowledge exists only in log files, inaccessible to future sessions.

2.2 Research Questions

RQ1: What memory architecture enables persistent learning across sessions while maintaining compatibility with the @-atom communication framework?

RQ2: Can the CRDT mathematics used for real-time dialogue convergence (Care Resolve) be extended to cross-temporal memory synchronization?

RQ3: Does memory injection improve coordination quality compared to memoryless baselines?

2.3 Contributions

1. **Four-layer memory taxonomy** (Episodic, Procedural, Semantic, Meta) with `@MEMORY.*` atoms extending DEUS.PROTOCOL v0.5.
2. **MEMORY-CRDT v1.1**: Formal specification for memory synchronization using utility-weighted LWW-Register CRDT with convergence guarantees.
3. **Hypercorrection discovery**: Global pre-session memory injection degrades LLM ensemble performance — a cognitive bias previously documented only in humans.

4. **CARE-ALERT system**: Gated real-time intervention achieving the first statistically significant improvement across all ARRIVAL experiments (Mann–Whitney $p = 0.042$).

3. Related Work

Retrieval-Augmented Generation (Lewis et al., 2020) [4] retrieves relevant documents to augment generation but operates at the document level without structured memory layers or cross-agent synchronization.

MemGPT (Packer et al., 2023) [5] introduces hierarchical memory management for LLMs with main context and external storage but targets single-agent scenarios.

Generative Agents (Park et al., 2023) [6] implement memory streams for simulated social agents with reflection and planning but focus on narrative coherence rather than structured coordination protocols.

MEANING-CRDT v1.1 [2] introduced CARE Resolve for real-time dialogue synchronization. MEMORY-CRDT extends this mathematical framework to the temporal dimension.

CRDTs (Shapiro et al., 2011) [8] provide formal guarantees for distributed data convergence. Our adaptation replaces physical timestamps with utility scores.

4. Architecture

4.1 Four Memory Layers

ARRIVAL-MNEMO organizes persistent knowledge into four layers:

Memory Layers

Episodic
: Session-level records: task, models, outcome, CARE score. TTL: 7–30 days. *"What happened."*

Procedural
: Validated strategies with success rates. Persistence requires $n_{\text{trials}} \geq 3$. *"How to act."*

Semantic
: Derived knowledge: rules, facts, calibration data. Persistence requires evidence_count $\geq$ 3. *"What is true."*

Meta
: Agent calibration: trust scores, domain-specific accuracy, behavioral patterns. Updated via exponential moving average. *"Who to trust."*

4.2 @MEMORY.* Atom Family

Nine new atoms extend DEUS.PROTOCOL v0.5:

Table 1: `@MEMORY.*` atom family.

Atom	Function
`@MEMORY.EPISODIC`	Reference/inject session-level memory
`@MEMORY.PROCEDURAL`	Reference/inject strategy-level memory
`@MEMORY.SEMANTIC`	Reference/inject knowledge-level memory
`@MEMORY.META`	Reference/inject calibration data
`@MEMORY.INJECT`	Request memory injection during session
`@MEMORY.STORE`	Request memory persistence after session
`@MEMORY.FORGET`	Flag memory for eviction
`@MEMORY.CONFLICT`	Signal contradicting memories
`@MEMORY.MERGE`	Request memory consolidation

4.3 Injection Protocol

Memory injection follows a seven-step pipeline:
1. **LOAD**: Read memory store from persistent storage (JSON).
2. **FILTER**: Select memories relevant to current `@GOAL` using keyword matching (MVP) or embedding similarity.
3. **FORMAT**: Convert to `@MEMORY.*` atom notation.
4. **INJECT**: Append to agent system prompts as `[MEMORY CONTEXT]` block.
5. **RUN**: Execute Arrival dialogue with memory-augmented agents.
6. **EXTRACT**: Create new Episodic memories; update Meta with calibration data.
7. **MERGE**: Integrate via MEMORY-CRDT.

5. MEMORY-CRDT v1.1

5.1 From Dialogue to Memory

MEANING-CRDT merges divergent agent positions within a dialogue:

$$\hat{\mathbf{v}} = \frac{\sum_i w_i \cdot \mathbf{v}_i}{\sum_i w_i}$$

MEMORY-CRDT applies the same principle across time. When two memory stores conflict, the merge operation selects the higher-utility version:

$$\hat{m} = \arg \max_{m \in \{m_1, m_2\}} \mu(m)$$

where $\mu(m)$ is the utility function:

$$\mu(m) = 0.25 \cdot \text{recency} + 0.30 \cdot \text{frequency} + 0.25 \cdot \text{quality} + 0.20 \cdot \text{validation}$$

5.2 Utility Components

- $\text{recency}(m) = \max(0,\ 1.0 - \text{age_days}/90)$. Linear decay over 90-day window.
- $\text{frequency}(m) = \min(1.0,\ \text{access_count}/N_{\text{layer}})$. N varies by layer: Episodic = 10, Procedural = 20, Semantic = 5, Meta = 10.
- $\text{quality}(m)$: layer-specific (care_resolve, success_rate, confidence, trust_score).
- $\text{validation}(m)$: evidence strength, layer-dependent.

5.3 CRDT Properties

MEMORY-CRDT Properties

The merge operation is a **Last-Writer-Wins Register** (LWW-Register) CRDT where the "timestamp" is replaced by utility score:

- **Commutativity**: $\text{MERGE}(S_1, S_2) = \text{MERGE}(S_2, S_1)$.
- **Associativity**: $\text{MERGE}(\text{MERGE}(S_1, S_2), S_3) = \text{MERGE}(S_1, \text{MERGE}(S_2, S_3))$.
- **Idempotency**: $\text{MERGE}(S, S) = S$.

5.4 Memory Debt

Analogous to Meaning Debt:

$$\text{MmD} = \frac{\sum_i \mu_i \cdot \text{conflict}(m_i)}{\sum_i \mu_i}$$

where $\text{conflict}(m) \in \{0.0, 0.5, 1.0\}$.

5.5 Convergence

Memory Convergence Theorem

Given a sequence of sessions $\{s_1, \ldots, s_n\}$ each producing memory updates, the memory store converges to a stable state as $n \to \infty$, provided: (1) the forgetting function removes memories with $\mu < \theta$; (2) merge resolves conflicts via utility maximization; (3) consolidation extracts patterns into semantic memories.

5.6 Relationship to MEANING-CRDT

Table 2: MEANING-CRDT vs. MEMORY-CRDT.

Property	MEANING-CRDT	MEMORY-CRDT
Domain	Dialogue positions	Memory entries
Temporal scope	Within a session	Across sessions
Weight function	Coherence	Utility
Merge granularity	Per-round	Per-session
Convergence target	Consensus	Consistent memory
Debt metric	Meaning Debt	Memory Debt
CRDT type	GCounter-like	LWW-Register

Unifying principle: Both CRDTs implement weighted convergence under conflict, applied at different temporal scales.

6. Pilot Experiment (Phase 14a)

6.1 Design

A controlled experiment with 10 MCQ questions across 5 domains. Models: DeepSeek V3, Llama 3.3 70B, Qwen 2.5 72B. Three conditions: Seed, Control (no memory), Treatment (memory injection).

6.2 Results

Table 3: Pilot experiment results.

Condition	Accuracy	Avg Care	Cost
Seed	3/5 (60%)	0.850	$0.008
Control (no memory)	4/5 (80%)	0.750	$0.008
Treatment (memory)	**5/5 (100%)**	**0.850**	$0.010

Memory injection corrected the one control error, improved CARE by +13.3%, and increased atom density by +8.6%. Total: $0.026.

7. Phase 14: Large-Scale Validation (Negative Result)

7.1 Design

Scaled to GPQA Diamond — 40 graduate-level science questions [7]. Alpha trio: GPT-4o + DeepSeek-V3 + Llama-3.3-70B-Instruct. Memory store: 5 episodic, 2 procedural, 3 meta. Keyword-based retrieval, top-8 injection, 800-char limit.

Data leakage guard: No correct answers, answer letters, question text, or choices stored.

7.2 Results

Phase 14 — Negative Result

Metric	Phase 13	Phase 14	Δ	p-value
Accuracy (N=35)	57.1%	51.4%	−5.7 pp	McNemar $p = 0.724$
CARE Resolve	0.966	0.916	−0.050	M-W $p = 0.985$
Meaning Debt	0.196	0.433	+0.237	M-W $p = 0.994$

Memory injection did not improve accuracy. The trend is negative but not statistically significant.

7.3 The Hypercorrection Effect

Analysis of auto-stored cognitive scars reveals the failure mechanism. When told "in past chemistry sessions, DeepSeekV3 was suppressed," GPT-4o begins second-guessing its own correct answers on ALL chemistry questions, not just the specific ones where suppression occurred. This parallels the well-documented human cognitive bias: pre-task warnings about past mistakes increase anxiety and worsen performance.

LLM ensembles empirically exhibit this classic human pattern: they need correction DURING the process, not warnings BEFORE it.

7.4 Implications

1. **Memory creation is solved**: Auto-logging perfectly captures who was suppressed, in which domain, and by how much.
2. **Memory delivery via global injection fails**: Static pre-session injection of generic memories degrades performance.

3. **Specific memory helps, generic memory hurts**: Seen questions improved (+20 pp); unseen degraded because keyword retrieval injected noise.

8. Phase 15: Gated CARE-ALERT Validation

8.1 Architecture

Phase 14's negative result motivates a paradigm shift: from global injection to **real-time intervention**.

CARE-ALERT: Monitor → Detect → Intervene

1. R1 (Alpha: GPT-4o) → clean prompt, no memory
2. R2 (Beta: DeepSeek-V3) → clean prompt, no memory
3. *Checkpoint #1*: MD > 0.5 ⇒ Level 1 `@CARE.ALERT`
4. R3 (Gamma: Llama-3.3) → prompt + alert (if fired)
5. *Checkpoint #2*: MD > 0.8 ⇒ Level 2 ESCALATED
6. R4 (Alpha: GPT-4o) → prompt + alert (if fired)

Alert design: Operational, not moralistic ("Beta selected C, address with `@EVIDENCE`" — NOT "you failed before"). Current state only. Never contains correct answers, historical data, or trust scores. Clean system prompt, identical to Phase 13 baseline.

8.2 Results: Unseen Questions (N=35)

Table 4: Three-phase comparison on unseen questions.

Metric	P13	P14	P15	Δ P13→P15
Accuracy	57.1%	51.4%	**57.1%**	+0.0 pp
CARE	0.966	0.916	**0.948**	−0.018
Meaning Debt	0.196	0.433	0.452	+0.256

Phase 15 restores accuracy to the Phase 13 baseline. The −5.7 pp degradation from global injection is fully eliminated.

8.3 Results: Seen Questions (N=5)

Table 5: Three-phase comparison on seen questions.

Metric	P13	P14	P15
Accuracy	20%	40%	**60%**
CARE	0.539	0.635	**0.746**
Meaning Debt	2.455	1.944	**1.602**

Monotonic improvement across all three metrics through all three phases.

8.4 Results: All 40 Questions

Table 6: Three-phase comparison on all 40 questions.

Metric	P13	P14	P15
Accuracy	52.5%	50.0%	**57.5%**
CARE	0.913	0.881	**0.923**

First Significant Result

CARE Resolve Phase 14 vs. Phase 15 (N=40):
Mann–Whitney $p = 0.042$ (significant at $\alpha = 0.05$).
This is the **first statistically significant result** across all ARRIVAL experiments.

8.5 Alert Statistics

Table 7: CARE-ALERT firing statistics.

Statistic	Value
Total questions	40
Questions with alert	7 (17.5%)
Accuracy with alert	57.1% (4/7)
Accuracy without alert	57.6% (19/33)
CARE with alert	0.968
CARE without alert	0.914

The alert does not harm accuracy. CARE is HIGHER when alerts fire (0.968 vs. 0.914).

8.6 Minority Voice Protection

Three examples of successful minority voice protection:

- **GPQA_009**: Beta (DeepSeek) correct, Alpha (GPT-4o) wrong. Alert fired → final answer followed Beta → CORRECT.
- **GPQA_038**: Gamma (Llama) correct, Alpha+Beta wrong. Alert fired → final answer followed Gamma → CORRECT.
- **GPQA_003**: Beta correct against Alpha. Alert → CORRECT.

8.7 Analysis

Finding 1: Gated alerts avoid hypercorrection. Phase 14 injected memory into 100% of questions → −5.7 pp. Phase 15 alerted on 17.5% → +0.0 pp.

Finding 2: First significant CARE improvement (Mann–Whitney $p = 0.042$).

Finding 3: Precision targeting works. 17.5% alert rate demonstrates intervention only when needed.

Finding 4: Memory delivery was the problem, not memory itself:

- Phase 13: Baseline (stateless) → 57.1%
- Phase 14: Global injection → 51.4% (−5.7 pp)
- Phase 15: Gated CARE-ALERT → 57.1% (baseline restored)

9. Discussion

9.1 The CRDT Unification

The most significant theoretical contribution is the unification of dialogue-level and memory-level synchronization under the same mathematical framework. Care Resolve operates at two temporal scales: **intra-session** (MEANING-CRDT: merging agent positions in real-time) and **inter-session** (MEMORY-CRDT: merging memory states across time).

This suggests a general principle: weighted convergence under conflict is a universal primitive for multi-agent coordination, applicable from millisecond-level token generation to month-level knowledge accumulation.

9.2 Cognitive Bias in LLM Ensembles

The hypercorrection discovery has implications beyond memory systems. If LLM ensembles exhibit human-like cognitive biases under pre-session priming, other known human biases may also manifest: anchoring effects (confirmed in Phases 20–21), groupthink, and confirmation bias.

9.3 Design Principles for AI Memory

1. **Create liberally, inject surgically**: Auto-log everything; inject only on detected divergence.
2. **Operational over moralistic**: Tell agents what IS happening, not what they SHOULD do.
3. **Current state only**: Never transmit historical failures or trust scores.
4. **Gated intervention**: Monitor → Detect → Intervene beats blanket pre-loading.

10. Limitations

1. $N = 40$ (35 unseen) — insufficient power for small effects.
2. Single agent trio tested — only Alpha configuration.
3. Non-significant primary accuracy result.
4. Keyword retrieval (MVP) — no semantic similarity.
5. No consolidation implemented (Episodic → Semantic).
6. MD thresholds (0.5, 0.8) set *a priori* without optimization.
7. Single run — no repeated measurements.
8. Cooperative-only scope — no adversarial memory poisoning.

11. Conclusion

Arrival-Mnemo extends the Arrival Protocol with persistent memory. The four-layer architecture provides structured knowledge organization while MEMORY-CRDT v1.1 ensures consistent synchronization using the same mathematics that powers real-time dialogue convergence.

The three-phase experimental progression reveals fundamental principles:

- **Phase 13** (Baseline): 57.1% on GPQA Diamond.
- **Phase 14** (Global injection): -5.7 pp degradation → hypercorrection discovered.
- **Phase 15** (Gated CARE-ALERT): Baseline restored; first significant Care improvement ($p = 0.042$); 17.5% alert rate.

Total project cost across all phases: \$1.17. The entire investigation — from architecture design through negative result discovery to validated real-time intervention — cost less than \$1.20.

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A. Memory Store Schema

Listing 1: Memory dataclass definitions.

```
@dataclass
class EpisodicMemory:
    id: str                 # Content-addressed hash
    session_id: str         # Source session identifier
    timestamp: str          # ISO 8601
    task: str               # Task description
    models: List[str]       # Participating models
    outcome: Dict           # {accuracy, care_resolve}
    care_resolve: float     # CARE Resolve score (0.0-1.0)
    meaning_debt: float     # Meaning Debt score
    key_insight: str        # Human-readable summary
    atoms_used: List        # @-atoms observed
    ttl_days: int = 30

@dataclass
class ProceduralMemory:
    id: str
    strategy_name: str
    task_type: str
    description: str
    success_rate: float
    n_trials: int
```

```
@dataclass
class SemanticMemory:
    id: str
    domain: str
    content: str
    confidence: float
    evidence_count: int

@dataclass
class MetaMemory:
    id: str
    agent_id: str
    trust_score: float
    domain_scores: Dict
    total_sessions: int
```

B. CARE-ALERT Format Example

Listing 2: Example CARE-ALERT injection.

```
@CARE.ALERT [Level 1 -- Divergence Detected]
After Round 2, agents hold different positions:
- Alpha (GPT-4o): selected answer C
- Beta (DeepSeek-V3): selected answer A
Interim Meaning Debt: 0.72

Please address divergence with @EVIDENCE and
@COUNTER. Consider whether Beta's position has
merit before finalizing.
```

C. MEMORY-CRDT Formal Properties

Commutativity: For any memories m_1, m_2 with the same ID: $\max(\mu(m_1), \mu(m_2)) = \max(\mu(m_2), \mu(m_1))$. □

Associativity: For any three stores A, B, C: $\text{merge}(\text{merge}(A, B), C) = \text{merge}(A, \text{merge}(B, C))$ because max is associative and set union is associative. □

Idempotency: For any store A: $\text{merge}(A, A) = A$ because for every memory $m \in A$, $\max(\mu(m), \mu(m)) = \mu(m)$, and $A \cup A = A$. □